How-To Library

Informatica

Enable Customer Managed Keys for MDM SaaS on Google Cloud

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Abstract

This article explains how to create your own native master encryption key for MDM SaaS on Google Cloud. You can control and maintain the key that you create. Use the key to control and restrict access to your organization's data. MDM SaaS leverages the keys that you control to encrypt data at rest, including data that persist in the data store, search indexes, and search index backup.

Supported Versions

- Informatica MDM SaaS May 2025 or later

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Overview

IDMC protects your organization's sensitive data in the cloud using organization-specific encryption keys that are generated and stored in the IDMC key management service (KMS). To prevent malicious access, the keys are encrypted using a master key that is stored in the cloud provider's KMS. The master key is provisioned in Informatica's Google Cloud account and varies by POD.

If you prefer, you can create a customer-managed key (CMK). When you create a CMK, you control access to it. However, you'll need to grant IDMC access to the CMK so that it can encrypt and decrypt your organization's sensitive data.

Creating a CMK offers the following benefits:

- You can restrict and control any access to your data.
- You can restrict the decryption of your data in emergency situations.
- You create and hold the key material in your KMS. The key is never exposed to your cloud service provider.
- You maintain full control of the key throughout its lifecycle. You can revoke access or delete the key at any time.

Note: When you create a CMK, your KMS and Informatica Intelligent Cloud Services POD must use the same cloud. For example, if your Informatica Intelligent Cloud Services POD is USW1 on Google Cloud, then you must store your CMK in Google Cloud.

You can find instructions related to Google Cloud in this article for reference purposes. For more detailed information about Google Cloud, see the Google Cloud documentation.

Step 1. Create a role in the Google Cloud console

In the Google Cloud console, create an access role for IDMC and give the role the appropriate cryptographic permissions.

1. Log in to the Google Cloud console.
2. In the **Search** bar, enter `Roles`.

   The **IAM & Admin Roles** page opens.
3. Click **Create Role**.
4. Set the role **Name** to `informatica-kms-access`.
5. Enter a **Title** and **Description**, and keep the default setting for **Role launch stage**.
6. Click **Add Permissions** and add the following permissions to the role:

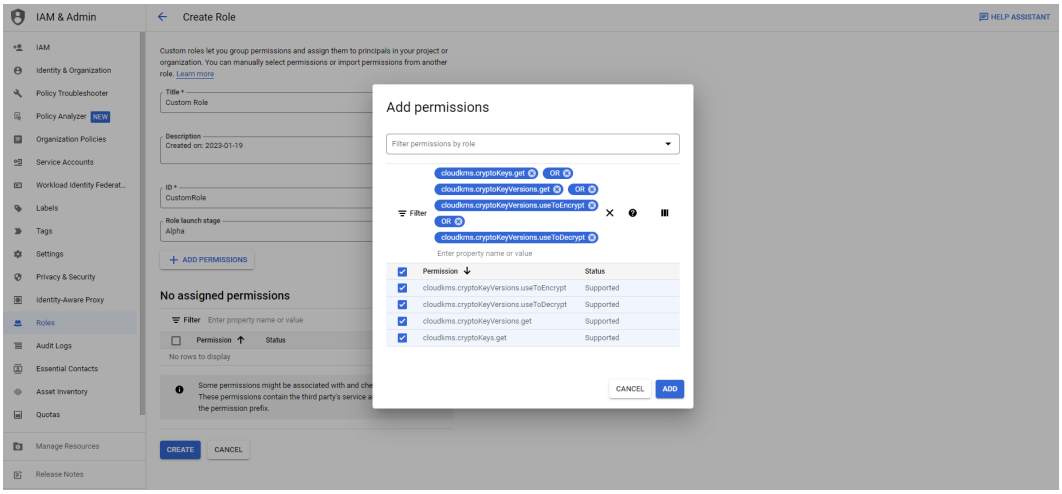
   cloudkms.cryptoKeyVersions.useToEncrypt

   cloudkms.cryptoKeyVersions.useToDecrypt

   cloudkms.cryptoKeyVersions.get

   cloudkms.cryptoKeys.get

   Tip: Use the **All Services** and **All Types** drop-down lists to filter and select permissions by services and types.



7. Click **Add**.
8. Click **Create**.

Step 2. Create the Google Cloud KMS key ring and key

Create three distinct encryption keys, consisting of one cross-region CMK and two single-region keys. The cross-region key secures data at rest within MongoDB Atlas and establishes cryptographic permissions for the IDMC service account. The two single-region keys secure Elasticsearch and backup storage buckets across the designated primary and disaster recovery regions.

1. In the Google Cloud console **Search** bar, enter `Key Management`.

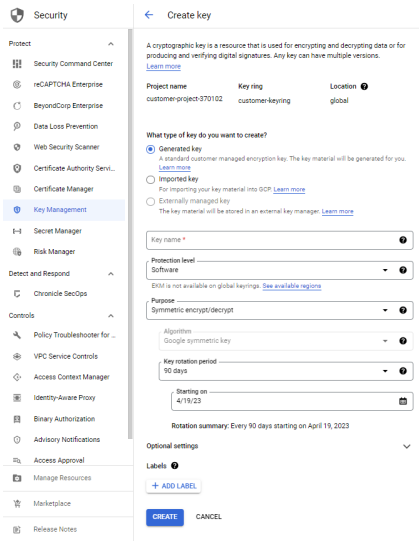
   The **Key Management** page opens.
2. Click **Create key ring**.

3. Configure the following properties for the key ring:

Property	Value
Key ring name	informatica-cmk-keyring
Location type	Region
Region	Select the appropriate region.

4. Click **Create**.
5. Select the key ring you just created.
6. Click **Create key**.
7. Configure the following properties for the key:

Property	Value
Key type	Generated key
Key name	Enter a meaningful name, for example, informatica-cmk-key.
Protection level	Software
Purpose	Symmetric encrypt/decrypt
Key rotation period	Set the key rotation to 90 days



8. Click **Create**.
9. Open the key and note the project, location, key ring name, key name, and key version.

Use the cross-region key for encryption within MongoDB Atlas and to authorize the IDMC service account. For more information, see "Step 5. Enable customer managed keys in MongoDB Atlas" on page 6 and "Step 3. Grant key access to Informatica's service account" on page 5.

Use the single-region keys to enable CMK in Elasticsearch for primary and disaster regions. For more information, see "Step 6. Enable customer managed keys in Elasticsearch" on page 8.

Step 3. Grant key access to Informatica's service account

Grant key access to Informatica's service account to give IDMC cryptographic permissions on the CMK. Informatica provides three principals.

1. In the Google Cloud console, select the key you created and open the **Permissions** tab.
2. Click **Grant Access** to select the permissions to grant to the key.
3. In the dialog box that appears, under **Add Principals**, add the principals obtained from Informatica.

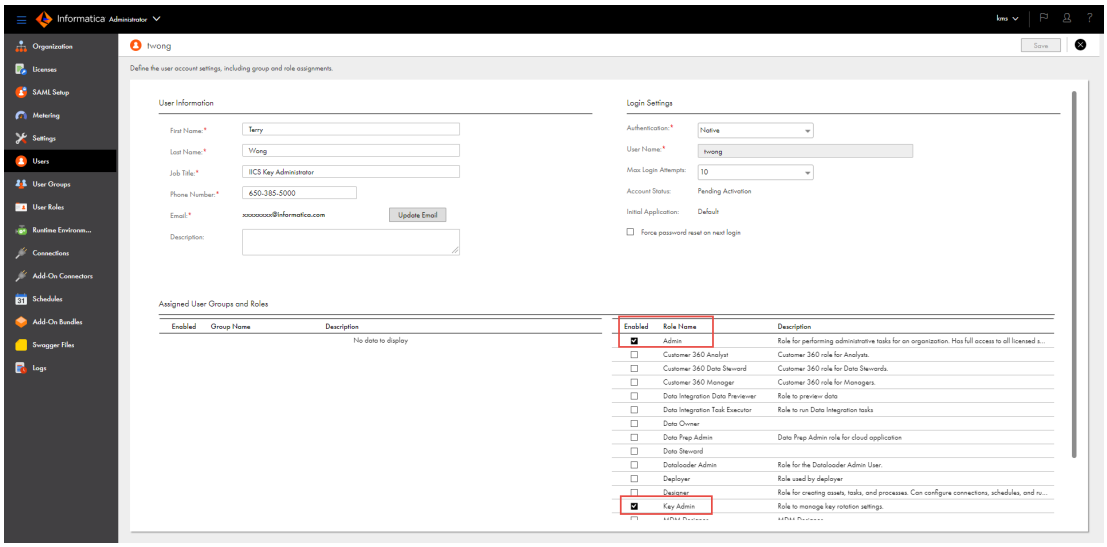
   An example of a service account principal obtained from Informatica is `svc-informatica-kms-connect@infa-cmk-test.iam.gserviceaccount.com`.
4. Under **Assign roles**, search for the role you created in "Step 1. Create a role in the Google Cloud console" on page 3.
5. Click **Save**.

   IDMC now has cryptographic permissions on the key.

Step 4. Enable customer managed keys in Informatica Intelligent Cloud Services

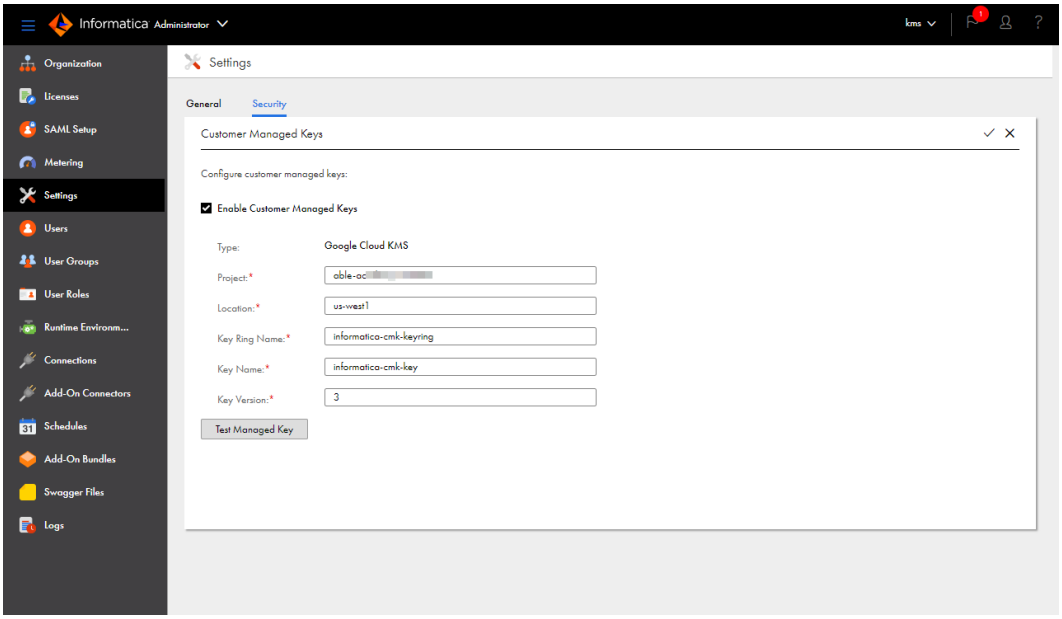
In IDMC Administrator, open the **Settings** page and enable customer managed keys for your organization.

Note: Before you can complete this step, you need to assign at least one administrative user the **Admin** and **Key Admin** roles on the user details page in Administrator:



1. Log in to IDMC Administrator with a user account that has both the Admin and Key Admin roles.
2. Open the **Settings** page and click the **Security** tab.
3. Click the edit (pencil) icon.
4. Enable the **Enable Customer Managed Keys** option.

5. Enter the **Project**, **Location**, **Key Ring Name**, **Key Name**, and **Key Version** for the CMK you created in "Step 2. Create the Google Cloud KMS key ring and key" on page 3:



6. Click **Test Managed Key** to test the key.

   A success message appears if the test was successful.

7. Click the save (checkmark) icon to save your changes.

   Note: It can take up to 24 hours for the key to become active.

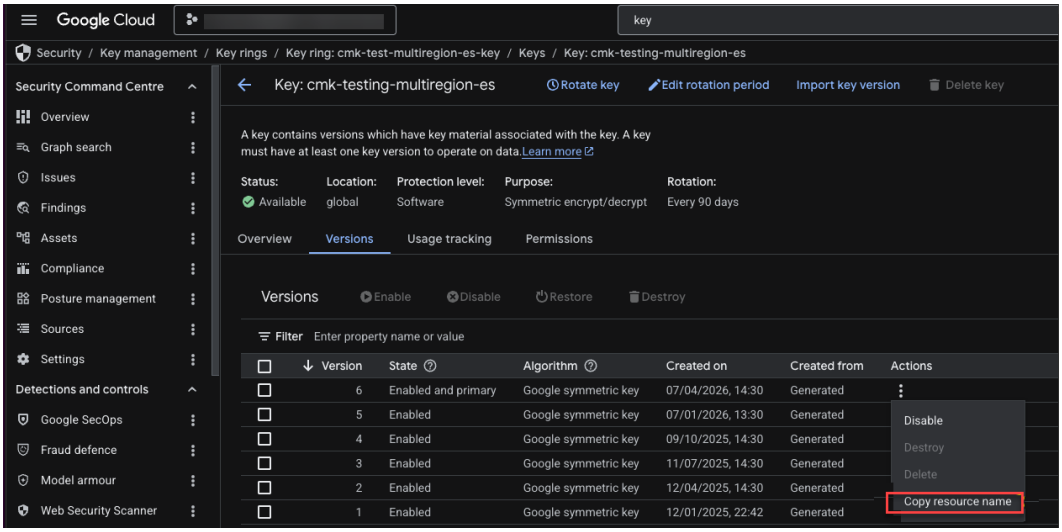
Step 5. Enable customer managed keys in MongoDB Atlas

To encrypt data at rest, link a CMK from your GCP KMS to your MongoDB Atlas. To enable encryption, contact Informatica Global Customer Support and provide your GCP key version resource name.

1. To obtain the resource name of the GCP key version, perform the following steps:
   a. In the Google Cloud console, go to the **Key Management** page, and select the key ring.
   b. Select the key that you want to use, locate the specific key version, and click **More**.
   c. Click **Copy resource name**.

   The following example shows a resource name:

`projects/iicsqa-mdmnext-cmktesting/locations/global/keyRings/cmk-test-multiregion-es-key/cryptoKeys/cmk-testing-multiregion-es/cryptoKeyVersions/6`

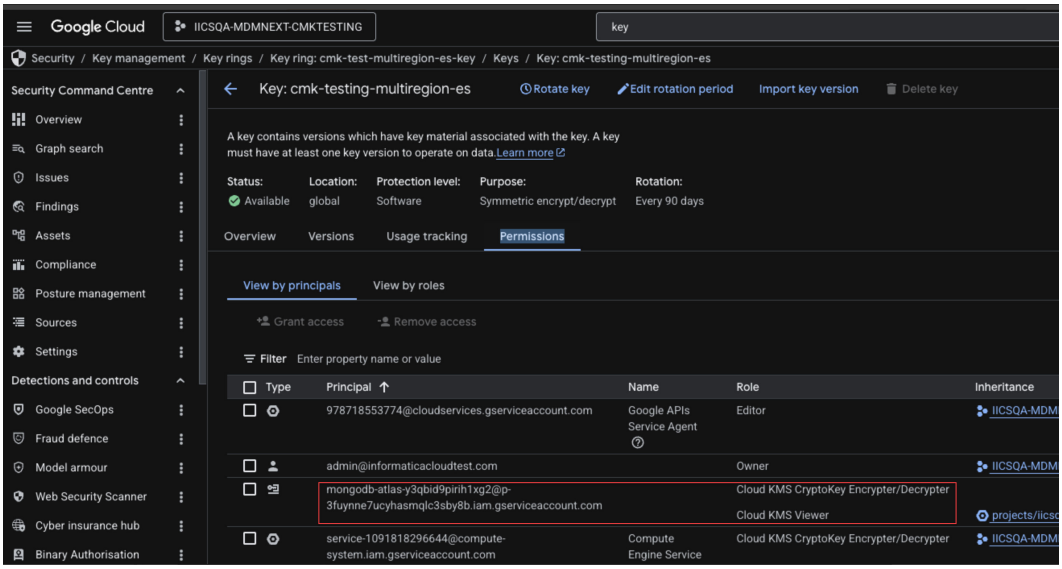


2. Run the commands provided by Informatica Global Customer Support in your terminal to grant the `Cloud KMS CryptoKey Encrypter/Decrypter` and `Cloud KMS Viewer` roles:

```
gcloud kms keys add-iam-policy-binding \
  cmk-testing-multiregion-es \
  --location=global \
  --keyring=cmk-test-multiregion-es-key \
  --member="serviceAccount:mongodb-atlas-
xu9uc4lvdqqu6ia9@p-3fuynne7ucyhasmqlc3sby8b.iam.gserviceaccount.com" \
  --role="roles/cloudkms.cryptoKeyEncrypterDecrypter"

gcloud kms keys add-iam-policy-binding \
  cmk-testing-multiregion-es \
  --location=global \
  --keyring=cmk-test-multiregion-es-key \
  --member="serviceAccount:mongodb-atlas-
xu9uc4lvdqqu6ia9@p-3fuynne7ucyhasmqlc3sby8b.iam.gserviceaccount.com" \
  --role="roles/cloudkms.viewer"
```

3. On the **Permissions** tab, verify that your MongoDB service account has the **Cloud KMS CryptoKey Encrypter/Decrypter** and **Cloud KMS Viewer** roles.



4. Notify Informatica Global Customer Support that you have verified that your MongoDB service account has the required roles.

   Informatica Global Customer Support encrypts your MongoDB service account and notifies you.

   For more information, see https://www.mongodb.com/docs/atlas/security-gcp-kms/#configure-your-authentication-method.

Step 6. Enable customer managed keys in Elasticsearch

Configure Elasticsearch encryption in both the primary and disaster recovery regions.

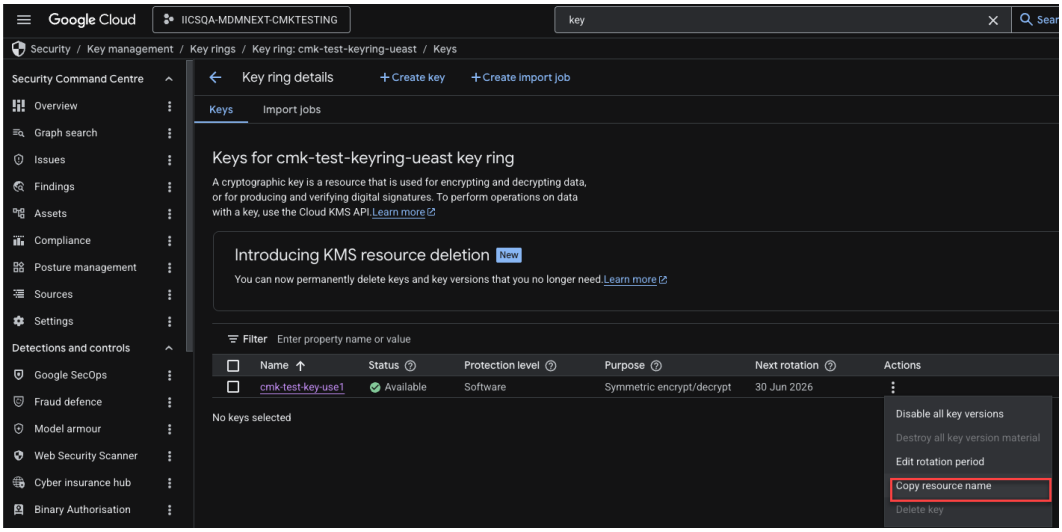
To complete the configuration, provide Informatica Global Customer Support with the two GCP KMS key ring resource names that you created in "Step 2. Create the Google Cloud KMS key ring and key" on page 3.

- To obtain the resource name of the GCP key ring key resource name, perform the following steps:
  a. In the Google Cloud console, go to the **Key Management** page, and select the key ring.
  b. Select the key that you want to use, and click **Actions**.
  c. Click **Copy resource name**.

  The following example shows a resource name:

  ```
  Primary region key resource name: projects/iicsqa-mdmnext-cmktesting/locations/us-west1/keyRings/cmk-test-keyring-elasticsearch-usw1/cryptoKeys/cmk-test-key-elasticsearch-usw1
  ```

Disaster region key resource name: projects/iicsqa-mdmnext-cmktesting/locations/us-east1/keyRings/cmk-test-keyring-ueast/cryptoKeys/cmk-test-key-use1



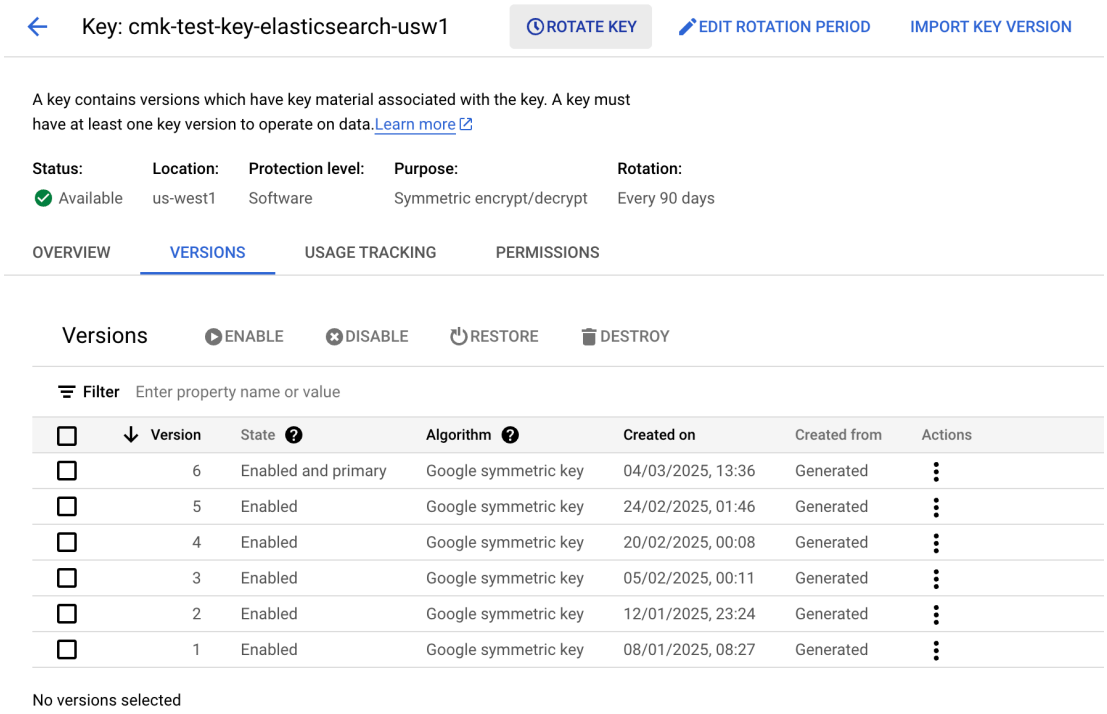
Informatica Global Customer Support secures the Elasticsearch instances, encrypts the specific storage buckets that contain the system backups, and notifies you.

Rotate the Key

You are responsible for tracking key expiration. To ensure operational efficiency and security, you must set the expiry to 90 days. Contact Informatica Global Customer Support every 90 days to provide the new version of the key to ensure MongoDB connects with the new secret of the key successfully.

Warning: If the key or secret expires, the MongoDB cluster becomes inaccessible, and MDM services are unavailable until recovery. Ensure that Informatica is provided with a new key and the new key is configured in the Mongo DB cluster at least 48 hours before the existing one expires.

When you rotate the key, you must keep all previous key versions in the Enabled state because some services depend on the old key versions for decryption. Services will fail if the keys are in the Disabled state.



Revoke access to KMS key

To restrict Informatica's access to your data temporarily, you can disable the key in Google Cloud KMS. After you disable the CMK, Informatica Cloud Services can't decrypt your organization's encrypted data, and any processes or jobs relying on this data will fail until you reactivate the CMK in Google Cloud KMS.

To revoke access to Elasticsearch data, contact Informatica Global Customer Support. The search indexes are stored on Persistent Disks, and the Persistent Disks volume retains the key in memory, which gets removed upon server reboot. The Informatica operations team manually shuts down the Elasticsearch cluster. Any subsequent operations fail if the key remains disabled.

Frequently asked questions

When I clicked **Test Managed Key** in IDMC, the test failed. What should I do?

If you get an error when testing the key, perform the following checks:

- In IDMC Administrator, verify that the key settings on the **Settings** page match the settings for the CMK in the Google Cloud console.
- In the Google Cloud console, verify that the status of the CMK is active.
- In the Google Cloud console, verify that the permissions on the CMK allow Informatica cryptographic access to the key.

If you continue to encounter errors, contact Informatica Global Customer Support.

What happens if the CMK is rotated in **Google Cloud KMS**?

You can rotate the CMK in Google Cloud KMS manually or on a schedule. Rotating a key creates a new version of the key. The old version of the key remains in Google Cloud KMS and is used for decryption only.

IDMC automatically detects key rotation. When the CMK is rotated in Google Cloud KMS, IDMC decrypts your organization's keys using the old CMK and then encrypts them using the new CMK.

What if I need to update the CMK in Google Cloud KMS?

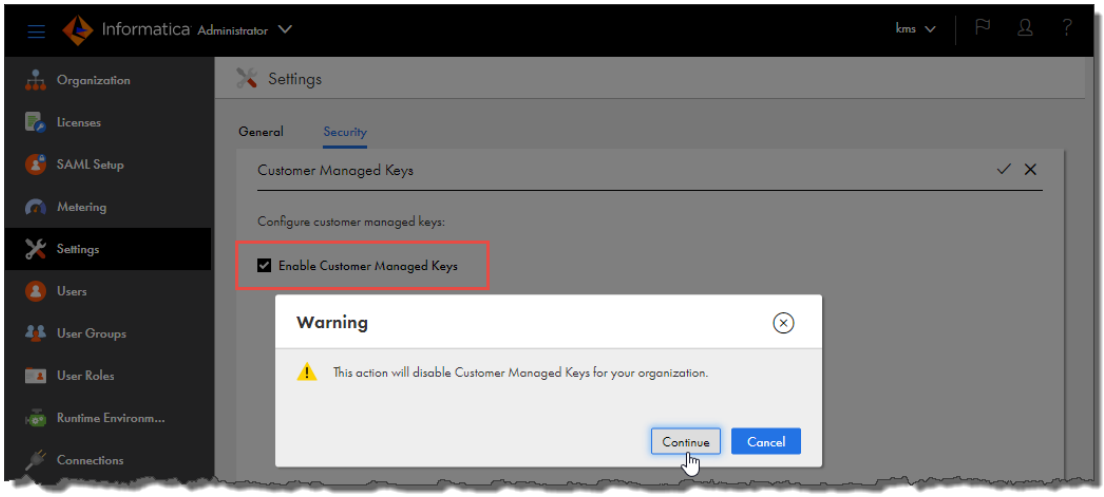
If you need to update the CMK, first provision a new CMK in Google Cloud KMS. Then, update the key details on the **Settings** page in IDMC Administrator.

Note: Be sure to keep the old version of the CMK in Google Cloud KMS active until you update the key details in IDMC.

You can delete the old version of the CMK in Google Cloud KMS after you update the key details on the **Settings** page in IDMC Administrator.

What if I want Informatica to manage key encryption?

If you want Informatica to manage key encryption, you can disable the **Enable Customer Managed Keys** option on the **Settings** page in IDMC Administrator:



When you do this, be sure to keep the current version of the CMK in Google Cloud KMS active. If the CMK is not active, disabling customer managed keys in IDMC fails.

When you disable this option, your organization's encryption keys are once again encrypted using encryption keys that are managed by Informatica. It can take up to 10 minutes for the Informatica encryption keys to become active.

You can disable or delete the CMK in Google Cloud KMS after you disable the **Enable Customer Managed Keys** option in Administrator.

What if I want to temporarily revoke Informatica's access to the CMK?

If you want to temporarily revoke Informatica's access to the CMK, you can disable the key in Google Cloud KMS.

When you disable the CMK, IDMC can no longer unencrypt your organization's encrypted data, and any jobs that use the data will fail until you reactivate the CMK in Google Cloud KMS.

How do I replace the CMK if I suspect it has been compromised?

If you want to replace the CMK, you can delete the key in Google Cloud KMS and create a new one.

Warning: Deleting the CMK in Google Cloud KMS results in permanent loss to any encrypted data in IDMC and causes the jobs that use the data to fail.

If you need to replace the CMK, perform the following steps so that you don't lose access to the encrypted data and jobs don't fail:

1. In Administrator, open the **Settings** page, click the **Security** tab, and disable the **Enable Customer Managed Keys** option.
2. In the Google Cloud console, delete the CMK.
3. In the Google Cloud console, create a new CMK.
4. On the **Settings** page in IDMC Administrator, re-enable the **Enable Customer Managed Keys** option and enter the details for the new CMK.

Can I delete the CMK if I don't want Informatica to access any of my encrypted data?

Warning: Deleting the CMK in Google Cloud KMS results in permanent loss to any encrypted data in IDMC and causes the jobs that use the data to fail.

If you're sure that you want Informatica to forgo all access to your encrypted data in IDMC, you can delete the CMK in Google Cloud KMS.

Can I use separate encryption keys for Development, Production, and QA organizations?

Informatica requires you to use a single master key for the deployed organizations. MDM SaaS leverages the master key to encrypt unique tenant-specific keys, ensuring that data isolation and security are maintained for each tenant individually.

Can I request my organizations to be deployed on different PODs?

When you use Customer Managed Keys, all organizations must be deployed within a single POD.

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